AI Tools in Algorithmic Management: Early Insights from Germany and Challenges for a fair and co-determined Design

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The increasing use of artificial intelligence (AI)-based systems in the workplace and their effects on workers are central topics in the current surge of AI. Especially in the field of algorithmic management (AM), the managing of workers based on or supported by algorithms. In this field the information and power imbalance between employees and employers can have a significant influence on the labor conditions. Moreover, it is often unclear what systems are actually being deployed, to what extent these include AI, and on what type of workers' data these systems operate. These questions are of utmost relevance to ensure a fair and co-determined design of AM processes. In our contribution we report early insights from an ongoing qualitative interview study conducted on the German labor market. We carried out semi-structured expert interviews with three works council representatives from companies in the electrical industry, IT, and healthcare sectors. Our preliminary findings indicate that AI-based AM is not yet widely spread in the more far-reaching areas of personnel management but expansions are already emerging. At the same time, challenges related to transparency, data governance, and co-determination are already becoming visible in practice. These results underline the importance of a socio-technical perspective on AI in AM and highlight the need for further research on both the technical design and governance structures of such systems to support fair and employee-centered implementation.

Keywords: Artificial Intelligence, Algorithmic Management, Labor, Digital Sovereignty

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1 Introduction

Algorithmic management (AM) originally referred to the practices in the platform economy, such as delivery and ride-hailing services, which rely on automated, algorithm-based management of the workforce [4]. In these contexts, for example, task allocation is automated and not directly supervised by a human manager. However, such practices have already spread to more conventional fields of work and are now taking on an even more significant role with the rise of artificial intelligence (AI) [1].
A recent study among managers found that 78% of those from German enterprises indicated that their company used AM to some extent [5]. This high level of adoption indicates that AM is no longer confined to platform-based

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work but is increasingly diffusing into more traditional organizational contexts. At the same time, the growing integration of artificial intelligence into AM systems raises important questions about its impact on employees. The use of AI in AM could have significant consequences for the well-being of workers ([3], [6], [7], [8]). In particular, AI-driven systems may influence workload allocation, performance evaluation, and decision-making processes in ways that are not always transparent or easily contestable. Due to the often opaque and complex nature of these systems, employees may find it difficult to understand how decisions affecting them are made. This opacity can further exacerbate existing power asymmetries between employers and employees, as managerial control becomes increasingly data-driven and automated. Thus, it is crucial to understand the use of AI in AM in order to devise measures for worker well-being under AM.

However, studies with deeper analyses of use cases, the types of algorithms, and the kind of data used in different fields of AM are rare.

To address this gap, our exploratory study aims to shed light on these questions. Drawing on qualitative interviews, we investigate how AI-based AM systems are currently used in practice. As our interview study is still ongoing, we present initial findings based on the first interviews conducted up to date. These early insights highlight emerging patterns in the use of AI in AM and point to key challenges, particularly with regard to ensuring ethical deployment and enabling meaningful forms of co-determination in AI-driven management contexts.

2 Methodology

To investigate the actual use of AI in AM, we conducted semi-structured expert interviews in early 2026. We focused on experts from larger companies with at least 1,000 employees, as such organizations were more likely to use AM. In this initial phase of our study we primarily targeted works council representatives, as they were relatively accessible and typically had a good overview of internal processes in their respective companies. On top of this, works council representatives in Germany are legally required to be involved in the introduction of technical measures that are intended to monitor employees' behavior and performance, which some forms of AM might. Of course the restriction of only having interviewed works council representatives limits the views and experiences represented in this early stage of the study as employees' experiences with AM may vary even within a single company. Inclusion of further stakeholders is planned for future phases of the study, e.g. employees and managers. Participants were recruited both individually and collectively, for example through contacts with unions.

All interviews were carried out remotely via video conference, with only the interviewer and one interviewee present. Each interview started with a brief introduction, followed by the question about the participant's understanding of AM to capture their initial perspective on the concept. After establishing this baseline, participants were asked some general questions about themselves and their company (e.g., their experience and the size of the company) to ease into the conversation.

The main part of the interviews was more open in structure. All participants were asked if they or colleagues were directly affected by AM and, if so, which software or algorithms were involved. Furthermore, we asked what types of employee data were used in these cases and how the data was collected. Additional questions addressed company guidelines on the use of AI and AM.

In total, eleven main questions were prepared to guide the interviews, some with follow-up questions to gather more detailed insights. We kept the definition of AI and AM throughout the interview very broad on purpose to identify as many use cases and systems as possible. The interviews lasted between 45 and 60 minutes.

The preliminary results presented here were based on the first three interviews conducted. The participants came from companies from electrical industry, IT, and the healthcare sector.

3 Preliminary results

Out of the three interviews, we found two main understandings of AM among the interviewees. Two experts identified AM as AI taking on lower management functions whereas one expert focused on the aspect of work simplification for colleagues via automation of auxiliary tasks.
Example use cases of AI in AM identified in our interviews included:

- the use of continuously trained, deep learning-based systems for task allocation, matching available and nearby employees to transportation tasks,
- the use of chatbots to guide employees through personnel management software,
- the use of language models to generate explanatory texts for otherwise automated task allocation decisions, and
- the use of agentic AI to support personnel management processes, for example by preparing relevant forms.

Other use cases of AI mentioned included the automated creation of employee skill profiles to guide the assignment of employees to tasks using large language models based on, e.g., chat data, the monitoring of physically demanding tasks to improve employees' ergonomics, and the automated pre-selection of relevant patient's data for employees in the healthcare sector, both based on computer vision. One interviewee also named the Microsoft Office suite and general Enterprise Resource Planning (ERP) tools as forms of AM employed in their company. Two out of three interviewees indicated that their companies were developing their own small language models for different tasks that partially fall under AM and that there were lots of ongoing developments of AI systems in their companies in general.
Two out of the three interviewees reported that keeping track of which employee data was used for which purposes and whether it was properly anonymised where required was not always feasible. They also noted that documenting the use of AI at a level that is both sufficiently detailed and generally understandable remains a significant challenge. This challenge mainly arose from the need for documentations to be both sufficiently technical to explain what kind of AI was being used and how it operated, and at the same time accessible to employees without AI expertise. As one interviewee stated, employees might be better able to judge such systems based on the types of data used rather than on the specific AI techniques involved.
Only one of the three interviewees reported that their company had a specific works agreement on the use of AI. In the other two cases, such agreements were envisioned for the future but currently the use of AI systems was regulated by more general works agreements on the processing of employee data and IT systems.
The interviews yielded the preliminary insight that the use of AI in AM is not yet widespread in more sensitive areas of personnel management, such as remuneration or promotions. They also suggested that the developments in this field have not been as quick as initially expected, at least within this small sample. However, the interviewees indicated that there were plans to expand the use of AI and AM into additional, and potentially more critical, areas of management. Figure 1 shows a generalized overview of the preliminary results from the interviews. Here, the results are grouped into use cases, tools, and challenges for the application of AI in AM.

Fig. 1. Visualization of the use cases, tools, and challenges for AI in AM mentioned by the experts.

4 Conclusion

Preliminary results from our expert interviews on the use of AI in AM in German companies suggest that the majority of current applications are not yet in the more far-reaching areas of personnel management, e.g. career management, as already indicated by [3]. However, the interviews also indicated that an expansion into more sensitive domains is already being planned. At the same time, initial challenges are becoming apparent.

One key issue concerns the development of co-determined works agreement on the use of AI, which appears to be a complex and potentially limiting factor for its fair and effective deployment in AM. In addition, ensuring that the collection and use of employee data comply with the GDPR [2] and national data protection laws remains a significant challenge in practice due to increased automated collection and evaluation of the data as also highlighted by [8]. This is illustrated by use cases such as the automated generation of skill profiles from company chat data, which can be perceived as particularly intrusive. These findings highlight that, even at an early stage, AI-based AM systems raise concerns related to transparency, data use, and employee participation. As outlined in the introduction, such issues are closely linked to questions of employee well-being and the risk of increasing existing power imbalances between employees and employers.

Against this background, further research is needed to better understand both the technical design and the socio-technical implications of AI tools in AM. This is essential for developing approaches that support fair, transparent, and employee-centered use of AI in the workplace.

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